Google Cloud

Data access management for secure data

Hello, data enthusiast!

I'm so happy you've joined me for this investigation into data access controls and best practices of data management.

As you're about to discover, a major part of data management is ensuring that data is secure.

And security, in this context, is about where the data is stored and who has access.

Let's begin with an example.

Picture an office manager with a security device at their office entryway.

They need to provide an up-to-date access code to their staff and trusted vendors.

In these circumstances, the office manager can give other people the access code to allow them in.

In other words, sharing this code means they can give access to only people they trust —and keep out those who shouldn't enter.

Chances are, you use unique identifiers like this access code all the time.

Other types of unique identifiers include the code to access a phone or a computer.

Data access management works in a similar way.

As a data professional, you give specific people access to your data and your data processes — and keep out those who shouldn't have access.

Ok, first a quick definition: Data access management is the process of implementing features including password protection, user permissions, and encryption to protect data and related processes.

In the context of the cloud, it includes the creation of individual user roles or user groups.

And the access for each role or group is pre-defined by an administrator.

The act of assigning this access is called identity access management, or IAM.

IAM is the process of assigning certain individual roles or groups with access to particular resources.

And some key resources may include software hosted in the cloud, datasets, databases, or entire data warehouses.

The level of access is determined by the role of the user.

There are three main components of identity access management: principal, role, and policy.

First, principal is the account or accounts that can access a resource.

You can think of this as the account attached to each username.

It's important that each account has a unique identifier.

The second aspect of identity access management is role.

This is the collection of permissions that lists the elements that an account has access to.

Each principal may have a different role.

And finally, policy is the collection of roles that are attached to a principal.

Let's explore how this might work in an organization.

As a data analyst administrator, you're setting up a security group for the entire data team in the Finance organization.

The principal is a group that contains all the individual usernames for those on the data team.

The roles in this group include both data writer and data reader.

Both roles are attached to the principal.

The policy is both data writer and data reader attached to the principal's username.

Now, universal access, access by role, and access by environment are the primary access types you'll encounter.

Here's how they work.

Universal access —as the name suggests— is access that everyone needs.

This may be universal access to an entire project or to one portion.

Next, access by roles is access for each individual role assigned to a project.

Creating profiles by roles helps determine what access is needed for the tasks each role will perform.

Finally, access by environment is determined by user location.

So, a remote user may have different access than someone who's onsite.

There are some best practices that help ensure data is accessed according to assigned permissions.

First, be sure to audit data access and monitor user tasks.

Second, put additional permissions on remote access.

Also, verify and stop unusual activities.

And finally, add two-factor authentication or strong passwords to accounts to ensure security.

It's also a good idea to regularly check access roles.

Users may not need as much access as originally provided.

Alternatively, a user may need increased permissions to complete a project.

And it's always possible that users change departments and take on new roles, and need an update to their permissions.

Along the way, it's very important to document user access so you have a record of who has access and why.

This also helps audits go smoothly and is critical for compliance documentation.

Consider sharing this information in a team playbook so access issues don't become blockers to completing work.

I hope you enjoyed this rundown of data access management!

With effective access policies in place, you'll know you're helping your employer keep its data safe and sound!